

The Climate Conversation Online: Analyzing Tweet Sentiments and Stances

Introduction

Social media plays a pivotal role in shaping society—from morals to fashion trends, everything is constantly influenced by the invisible grasp that social platforms have on our lives. Given this, social media activity can reveal a significant amount about oneself. On X, a website formerly known for its brief 140 character limit and by the name Twitter, users openly share public rants, political opinions, personal stories, feelings, and more (Pardes). This project explores applying Natural Language Processing to evaluate public opinions on climate change, an increasingly pressing societal issue, through tweets. Specifically, it explores what climate change-related tweets reveal about sentiments and stances on the topic.

I hypothesize that climate change tweets of positive sentiment tend to correlate with a “believer” stance, whereas negative climate change tweets are related to a “denier” stance. When people believe in climate change, they likely want to take concrete steps towards preventing its disastrous effects, since there is a risk of irreversible damage being done to the planet. Believers think we need to take action as soon as possible—hence the prediction that, on average, their tweets will have more of a positive sentiment, motivating others to take part in change. On the other hand, deniers often claim climate change is a “hoax” that is diverting attention from other “more important” issues that better deserve our energy and political and financial resources. This belief demonstrates pessimism, resulting in the prediction that deniers tweet about climate change with a negative sentiment. A study done in 2023 found that areas with more deniers tended to tweet with a negative sentiment and that there was a 90% correlation between sentiment and stance, which further supports this hypothesis (Effrosynidis et al). For my project,

I recreated similar results with a different dataset, supporting the idea that sentiment and stance are correlated.

In order to lessen the effects of climate change, significant societal change is needed. This requires (1) converting people into believers and (2) inspiring them to take action. Assuming that sentiment and stance are correlated, climate activists can use this information and its implications to their advantage in an attempt to convince people to take action. Furthermore, by exploring the polarization in the climate change conversation and how deniers approach such discussions, advocates can better understand both the audience that they are attempting to reach and how to message towards them more effectively.

To address the hypothesis, I first classified tweets by stances, attempting to match the original labels. While this was not very successful, and only resulted in an accuracy of about 40%, this could be due to largely imbalanced data, which is addressed in a later section. Next, I applied a model fine-tuned on tweets to assign a positive, neutral, or negative sentiment to the tweets. Using the model's results, I explored the relationship between sentiment and stance. I calculated a p-value of $7.78e-119$ and performed post-hoc analyses, indicating that correlation existed between stance and sentiment, but it was not clear how, since all pairwise p-values were below 0.05. Inspired by Effrosynidis et al's study, I graphed average stance with sentiment and found a correlation where average stance was higher for positive sentiment tweets (more likely to be believers). To obtain the most accurate comparison with the original paper, I compared my results with those of their dataset with labeled stance and sentiments. The trend in average stance across sentiments from my dataset and the original dataset followed a similar pattern when comparing overall trends in the two graphs—positive sentiments had a higher average stance (where believers are 1, neutrals are 0, and deniers are -1).

Exploratory Analysis and Preprocessing

Before applying any rules or models, I explored the data makeup and preprocessed the text. My overall findings were that of the 31,960 tweets (after removing duplicates and news), around 65% of the tweets were believers, 24% were neutral, and only 12% were deniers. This data makeup is important when considering the results of the models later.

For preprocessing, I converted all tweets to lowercase and removed URLs, usernames, special characters, and digits. For example, before preprocessing the first tweet was ‘RT @NatGeoChannel: Watch #BeforeTheFlood right here, as @LeoDiCaprio travels the world to tackle climate change <https://t.co/LkDehj3tNn> <http://t.co/LkDehj3tNn>.’ After, it was ‘rt watch beforetheflood right here as travels the world to tackle climate change <http://t.co/LkDehj3tNn>.’ This step is essential so that the model only considers the most relevant, understandable words when processing the tweets.

Stance Analysis

In an attempt to replicate the stances given in the dataset, I first removed stop words such as ‘climate,’ ‘change,’ and ‘rt,’ that seemed to be in the top-n-words for all stances. Because such words were prevalent in tweets regardless of stance category, they would not assist much in distinguishing between the stances. I also used the built in stopwords list. Then, I used CountVectorizer to help tally up the counts of each word in each category to find the 15 most common words in the believer, neutral, and denier tweets. I handpicked from these the unique words in each category and added my own keywords to the rule-based lists. After assigning stances, I made a confusion matrix to see how successful the rules-based classification was.

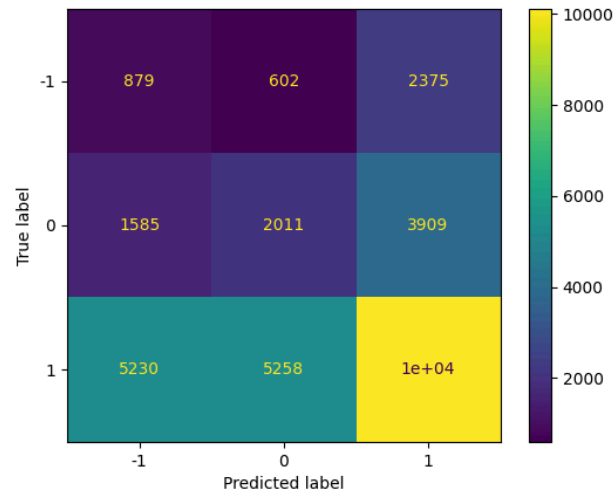


Figure 1. Predicted vs True Stances

As seen in **Figure 1**, the rules-based classification worked fairly well for predicting a tweet by a believer, but not as much for the others. This follows the numerical findings, which indicated only a 41% accuracy score, but a high precision of 62% for believer tweets. This finding was to be expected given that the majority of the dataset consisted of believer tweets, so there was more data to parse common words from. So while I was relatively successful in assigning believer tweets the correct stance, I only had a small sample size of tweets to analyze for neutral and denier stances, making it harder to see what words are typically common with such stances. This is further shown by the top left four corners of the graph being dark purple, meaning not many predictions were correct for neutral and denier tweets. This is potentially what is bringing down the average accuracy to 41%. Given the time constraints of this final project coupled with the fact that this was not the main focus of my project, I did not expand further on my lists (although I did increase accuracy from the original 23%). Given more time, I would dedicate more effort towards preprocessing such as stemming word endings (such as -ing, -s) and removing more custom stopwords. Another way to potentially improve results is to use a custom

model approach to predict stance. I could also acquire more neutral and denier data so the imbalance is less pronounced and there is more data to learn from.

Sentiment Assignment

For the sentiment assignment, I used a Hugging Face model that was fine-tuned for usage on tweets, called the “Twitter Roberta Base Sentiment Model” (linked in bibliography). Below are some examples of the classifications.

Positive	‘fabulous leonardo dicaprio s film on climate change is brilliant do watch via’
Neutral	‘climate change is an interesting hustle as it was global warming but the planet stopped warming for yes while the suv boom’
Negative	‘arseholes couldn t predict tomorrows weather in a month heat wave yet we trust them on climate change narrnegative’

Table 1. Examples of Tweets for Each Sentiment

Sentiment vs Stance: New Data

To explore correlation between sentiment and stance and address my hypothesis, I used the original stances from the data and the sentiments from my above analysis to calculate a Chi-squared value (555) and p-value ($7.8e-119$). The low p-value indicates that the relationships present are not due to chance and that there is a statistical significance between stances and sentiments. To explore this further, I conducted pairwise tests. However, these were not very conclusive since all the p-values were very low. Some of the more significant relationships (based on Chi-squared values) were Negative, Neutral and Deniers, Neutral and Negative, Neutral and Deniers, Believers. To find out which direction the correlation goes in, I was inspired by the 2023 study to compare average stance across the three sentiments (Effrosynidis et al). **Figure 2** below shows that negative tweets had a low average stance while positive tweets had a very high average stance across the three sentiment categories. Since I used the average

stances, I know this is not solely due to the sheer volume of believer tweets. This suggests that positive sentiment tweets are correlated with being written by believers, and negative sentiment tweets are correlated with being written by deniers. This is in line with my hypothesis.

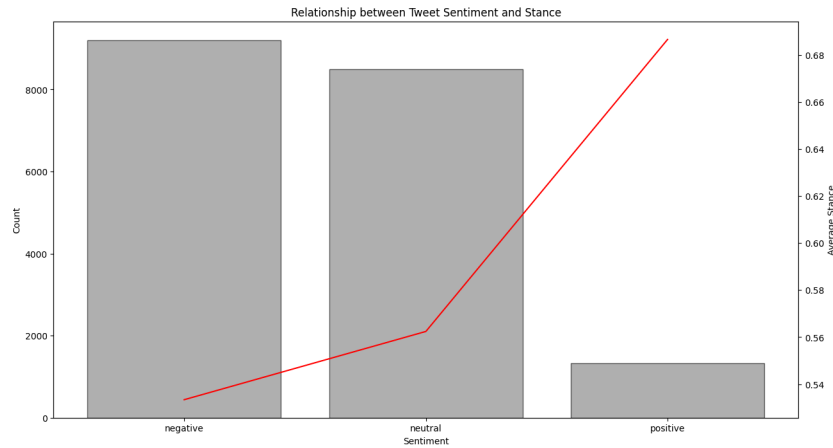


Figure 2. Average Stance across Sentiments I

Sentiment vs Stance: Past Research

Since one purpose of my project was to replicate the findings of the paper I was inspired by, I decided to also analyze their data and replicate their graph for comparison purposes. There was a very low p-value (0) again, indicating a statistically significant difference in sentiments across the stance categories.

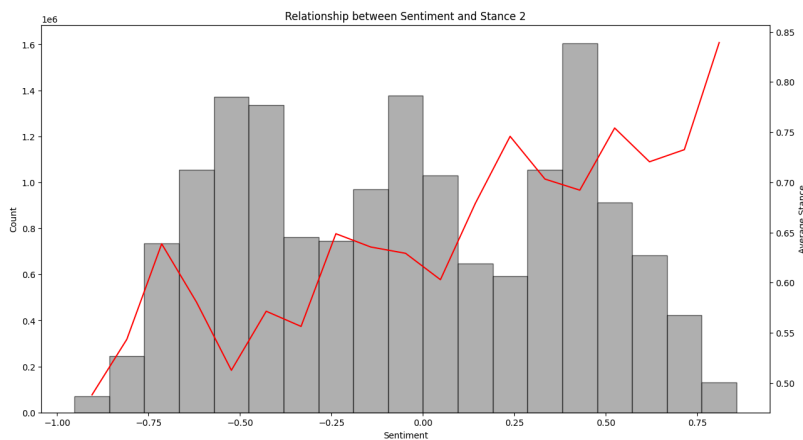


Figure 3. Average Stance across Sentiments II

Again, one sees the expected correlation between sentiment and stance. Since this data had sentiment as a continuous variable, the correlation is even clearer. As sentiment goes up (more positive) so does the average stance (towards believers). The direction of the average stance (red line) is similar in my graph, indicating that I was able to replicate similar findings, to an extent. My results are slightly different since sentiment was a categorical variable rather than a continuous one. Since my average stance line matches this and is in line with the direction suggested by my hypothesis, this further supports my project's replication and hypothesis.

Findings

Overall, my main finding is that there is a correlation between sentiment and stance in climate change tweets. Believers tend to tweet more positively, while deniers tend to tweet more negatively. Upon further examination of the top 15 unique words in each category's tweets, this finding is logical. For example, believers' tweets commonly say 'action' and 'epa,' which carry positive connotations linked to proactiveness and beneficial change. On the other hand, deniers' tweets' commonly read 'scam,' 'fake,' 'obama,' and 'liberals.' These words are associated with skepticism, distrust, and political polarization—all of which have negative sentiments. Since my project was able to find similar results to Effrosynidis et. al's paper, this further supports their proposed idea that there is a correlation between tweet sentiment and its author's stance on climate change. The findings of my project and their paper are key when considering how to inspire people to take action against climate change and believe in climate action's importance. Now knowing that there is a correlation between negative sentiment and denying climate change, climate activists can target the skepticism felt by deniers directly by countering misconceptions, addressing fears, and explaining why climate action is beneficial for all—ultimately aiming to convert deniers into believers and inspire them to take action.

Bibliography

Effrosynidis D, Sylaios G, Arampatzis A (2022) Exploring climate change on Twitter using seven aspects: Stance, sentiment, aggressiveness, temperature, gender, topics, and disasters. PLoS ONE 17(9): e0274213. <https://doi.org/10.1371/journal.pone.0274213>

Pardes, Arielle. "A Brief History of the Ever-Expanding Tweet." *Wired*, 26 Sep 2017, <https://www.wired.com/story/a-brief-history-of-the-ever-expanding-tweet/>. Accessed 11 Dec 2024.

Dataset

<https://www.kaggle.com/datasets/edqian/twitter-climate-change-sentiment-dataset>

Hugging Face Model

<https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>

<https://huggingface.co/blog/sentiment-analysis-twitter>

Colab Notebook

<https://colab.research.google.com/drive/1tY8KvuT5OWbDXB6E3U-GEFFHjACqlnnE?usp=sharing>